



UDP Migration Spec: Manifest Visuals

Exactly what moves from PowerReact into the UDP monorepo to support the visual library, the gallery and the manifest reader.

SPEC VERSION 1.0 · 2026-10-08	SOURCE REPOSITORY PowerReact (branch PBI-React-Implementation)	TARGET REPOSITORY udp-monorepo-development (pnpm workspace)
ITEMS 48 items in 4 workstreams and 4 phases	MACHINE-READABLE TWIN <code>docs/udp-migration/udp-migration-manifest.json</code>	COMPANION Manifest Visuals: Architecture and UDP Integration (team document)

0 Instructions for the implementing agent

Read this page first. Every item below carries an ID, a phase, an action, a source and a destination; follow them literally.

1. Work phase by phase (P1 → P4). Stop at each gate (section 10) and run its checks; do not start the next phase while a check fails.
2. Paths use the variables of section 2: sources are relative to the PowerReact root, destinations to the UDP monorepo root.
3. Apply each item’s action exactly as defined in section 3. **ADAPT** changes only what its notes list; logic, names and tests stay.
4. Keep every invariant of section 8. If an item seems to require breaking one, stop and ask.
5. Items of workstream W4 (UDP API) are specifications for the UDP API team: do not edit that repository.
6. Use pnpm in the UDP monorepo (Stencil 4.41, React 18). Never copy `.env` files, credentials or tenant data.
7. Open decisions (section 11) have a proposed answer; use it unless the UDP team has decided otherwise, and say which you used.
8. The JSON twin (`docs/udp-migration/udp-migration-manifest.json`) lists the same items; if the two ever differ, stop and report it.

1 Scope at a glance

Four workstreams move the visual library, the manifest reader and their assets into UDP; the fourth is new server work in the UDP API.

25

web components (75 files)

48

move items

20

dependency entries

4

phases, 3 gates

WS	WORKSTREAM	DESTINATION	ITEMS	FILES
W1	Visual library	UDP Stencil library, generated wrappers and enterprise wrappers \$STENCIL · \$WRAP_PKG · \$ENT/UI/powerbi-visuals	19	176
W2	Manifest engine and reader	Enterprise React library: engine, renderer, pages and widget \$MANIFEST · \$ENT/udp/pages · \$ENT/UI/widgets	21	38

WS	WORKSTREAM	DESTINATION	ITEMS	FILES
W3	Assets and scripts	Map boundaries, seed manifests, JSON Schema, optional Univerus theme \$STATIC/geo · \$MANIFEST/___fixtures___ · scripts	5	14
W4	UDP API (outside the monorepo)	Query endpoint and manifest store, built by the UDP API team UDP API · ConfigService	3	3

2 Path variables

Expand these before acting. \$STENCIL mirrors \$PKG one level down, so relative imports and styleUrls keep resolving.

VARIABLE	PATH	REPO	WHAT IT IS
\$PKG	packages/udp-powerbi-visuals/src	PowerReact	The visual library source
\$APP	src	PowerReact	The PowerReact app source
\$STENCIL	packages/udp-stencil-component-library/src/components/powerbi-visuals	UDP	New folder; mirrors \$PKG one level down
\$STENCIL_PKG	packages/udp-stencil-component-library	UDP	UDP Stencil library root (@univerus/udp-stencil-component-library)
\$WRAP_PKG	packages/udp-react-stencil-component-library	UDP	Generated React wrappers (@univerus/udp-react-stencil-component-library)
\$ENT	packages/udp-react-enterprise-component-library/src	UDP	Enterprise React library source (@univerus/udp-react-enterprise-component-library)
\$MANIFEST	packages/udp-react-enterprise-component-library/src/manifest	UDP	New folder for the manifest engine and reader
\$STATIC	public	UDP app	The host app's static assets folder (served from the app origin)

3 Actions

ACTION	MEANING
COPY	Copy byte for byte. Change only the import paths the item lists.
EXTRACT	Copy only the named symbols into the new file; leave the rest of the source behind.
ADAPT	Port the file; replace only the imports and components the item lists. Keep logic, names and tests.
NEW	Create in UDP following the pattern given; nothing equivalent exists in PowerReact.
REGENERATE	Do not copy: the UDP build produces it.
REPLACE	Do not move: use the named UDP component or service instead.

ACTION	MEANING
REFERENCE	Do not move: use it as the specification for code the UDP team writes elsewhere.
EXCLUDE	Never move.

4 W1 • Visual library

19 items · 176 files · UDP Stencil library, generated wrappers and enterprise wrappers.

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
V01 P1	COPY	\$PKG/components/kpi/udp-pbi-*/	\$STENCIL/components/kpi/	5 components: kpi-card, kpi-hero, kpi-trend, kpi-bullet, kpi-variance (tsx, css, readme.md each). Keep the mirrored layout: relative imports <code>../../../../styles utils functional</code> must resolve to \$STENCIL. 15 files
V02 P1	COPY	\$PKG/components/charts/udp-pbi-*/	\$STENCIL/components/charts/	18 components: boxplot, bullet-bars, calendar-heatmap, choropleth, column-chart, diverging-bars, donut, dot-plot, ibcs-variance, point-map, ranking-bars, scatter, spotlight-bars, stacked-bars, timeline, treemap, trend-chart, waterfall. 54 files
V03 P1	COPY	\$PKG/components/tables/udp-pbi-*/	\$STENCIL/components/tables/	2 components: data-table, matrix. 6 files
V04 P1	COPY	\$PKG/functional/*.tsx	\$STENCIL/functional/	chart-kit, frame, icon, kpi-shell, map-kit, meter, states. 7 files
V05 P1	COPY	\$PKG/utils/** (non-test)	\$STENCIL/utils/	Root (data, events, formats, icons, palette, roving, series-chart, width-observer), export/ (csv, download, xlsx), layout/ (17 modules incl. geo, kpi), table/ (builders, model). 30 files
V06 P1	COPY	\$PKG/styles/*.css	\$STENCIL/styles/	tokens-bridge, motion, shadow, surfaces, charts, kpi, maps. tokens-bridge.css stays the only file that reads <code>--u-*</code>. 7 files
V07 P1	ADAPT	\$PKG/utils/**/*.test.ts	\$STENCIL/utils/ (same folders)	Vitest → Stencil Jest (<code>stencil test --spec</code>): drop <code>import ... from 'vitest'</code> (globals). vocabulary-e.test.ts reads <code>public/geo/*.json</code>: point it at the copy made in A01 (or a test fixture). 8 files
V08 P1	REGENERATE	\$PKG/components.d.ts	\$STENCIL_PKG/src/components.d.ts	UDP's Stencil build writes its own; do not copy. 1 file
V09 P1	ADAPT	\$PKG/index.ts	\$STENCIL_PKG public type entry	Add its type re-exports (events, formats, table model, KpiIconName, VisualFrameMode, BarMeta) to the library's public types. 1 file
V10 P1	ADAPT	packages/udp-powerbi-visuals/package.json	\$STENCIL_PKG/package.json	Merge dependencies per section 9. Keep UDP's xlsx 0.18.5 (do not add the 0.20 tarball). 1 file

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
V11 P1	ADAPT	\$PKG/utils/icons.ts	\$STENCIL/utils/ icons.ts	Uses lucide (38 icons). Either add lucide (section 9) or map each icon to UDP's icon registry; keep the exported names (KPI_ICONS, UI_ICONS, KpiIconName) — the manifest schema lists the KPI icon names. 1 file
V12 P1	COPY	\$APP/components/powerbi- visuals/__fixtures__/ samples.ts	\$STENCIL/__fixtures__/ samples.ts	Sample data (illustrative, no client data). Rewrite the type import @powerreact/udp-powerbi-visuals → relative imports of the same types. 1 file
V13 P1	NEW	\$APP/pages/Gallery.tsx (reference for content)	\$STENCIL/stories/ <family>.stories.ts	Replaces the Gallery page: one Storybook file per family (KPI, Ranking, Magnitude, Change over time, Deviation, Part-to-whole, Flow, Distribution, Correlation, Spatial, Exact values) using V12 data; a toolbar toggle for UDP dark (body.dark-theme) and the Univerus look (A05). 11 files
V14 P1	ADAPT	\$APP/components/powerbi- visuals/ elements.ssr.test.ts	\$STENCIL/__tests__/ elements.spec.tsx	Server-render smoke tests (full, empty, frame="none") for all 25 components; port to Stencil newSpecPage or keep the hydrate renderer. 1 file
V15 P1	ADAPT	\$APP/theme/ tokensBridge.test.ts + package part of \$APP/theme/ noRowColors.test.ts	\$STENCIL/__tests__/ tokens.spec.ts	Guards: every --pbi-* read exists in the bridge with a Fluent fallback; no var(--u- outside tokens-bridge.css; no hex / rgb / hsl literals in \$STENCIL. 2 files
V16 P1	ADAPT	packages/udp-powerbi- visuals/README.md, MIGRATION.md	packages/udp-react- enterprise-component- library/docs/charts/ powerbi-visuals.md	Component table, contract, token bridge, maps, the documented frame="card" deviation. 2 files
R01 P1	REGENERATE	\$APP/components/powerbi- visuals/generated/ components.ts	\$WRAP_PKG	Built by UDP's React output target (React bundle). Never copy PowerReact's generated file. 1 file
R02 P1	NEW	—	\$ENT/UI/powerbi- visuals/ UdpPbi<Name>.tsx + index.ts	One thin wrapper per component (25) + barrel, the pattern of UDP's chart wrappers; export the barrel from \$ENT/index.ts. 26 files
R03 P1	ADAPT	\$APP/components/powerbi- visuals/index.ts	\$ENT/UI/powerbi- visuals/index.ts	Public type re-exports used by the manifest code (DataPointClickDetail, DataPointValue, GeoPoint, RegionValue, KpiPoint, ...), now from the UDP wrapper package. 1 file

R02 · wrapper pattern

```
// $ENT/UI/powerbi-visuals/UdpPbiDonut.tsx - one per component (25)
import React from 'react';
import { UdpPbiDonut as StencilUdpPbiDonut, type JSX } from '@univerus/udp-react-stencil-component-library';

export const UdpPbiDonut: React.FC<Partial<JSX.UdpPbiDonut>> = (props) => <StencilUdpPbiDonut { ... props} />;
```

5 W2 · Manifest engine and reader

21 items · 38 files · Enterprise React library: engine, renderer, pages and widget.

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
M01 P3	COPY	\$APP/lib/manifest/schema.ts	\$MANIFEST/lib/schema.ts	Zod 4 schema: 26 components, schemaVersion: 1 , strict objects. 1 file
M02 P3	COPY	\$APP/lib/manifest/jsonSchema.ts	\$MANIFEST/lib/jsonSchema.ts	Set MANIFEST_SCHEMA_FILE to the UDP location of the published JSON Schema (A04). 1 file
M03 P3	COPY	\$APP/lib/manifest/daxInjection.ts	\$MANIFEST/lib/daxInjection.ts	Import <code>../dax/daxBuilder</code> → <code>../dax</code> (M07). The only place filters enter a query. 1 file
M04 P3	COPY	\$APP/lib/manifest/mapRows.ts	\$MANIFEST/lib/mapRows.ts	Imports: <code>@powerreact/udp-powerbi-visuals</code> (types) → <code>@univerus/udp-react-stencil-component-library</code> ; <code>../dax/parse</code> , <code>../dax/types</code> → <code>../dax</code> . 1 file
M05 P3	COPY	\$APP/lib/manifest/crossFilters.ts	\$MANIFEST/lib/crossFilters.ts	Same import rewrites as M04. Holds Sample mode and which filters each visual obeys. 1 file
M06 P3	ADAPT	\$APP/lib/manifest/*.test.ts + \$APP/components/manifest/registry.test.ts	\$MANIFEST/__tests__	Needs a unit-test runner in the enterprise library (decision D1). Bundled manifests (A03) as fixtures. 7 files
M07 P3	EXTRACT	\$APP/lib/dax/daxBuilder.ts, parse.ts, types.ts	\$MANIFEST/lib/dax.ts	Only: <code>daxString</code> , <code>DaxLiteralValue</code> , <code>daxLiteral</code> , <code>treatAs</code> , <code>call</code> (<code>daxBuilder</code>); <code>toNumber</code> , <code>toLabel</code> (<code>parse</code>); <code>DaxRow</code> (types) — with their test cases. Not the asset-page builders, <code>GROUP_COLUMN</code> or <code>CLASS_COLUMN</code> . 1 file
M08 P3	EXTRACT	\$APP/store/filters.ts	\$MANIFEST/store/manifestFilters.ts	Only the cross-filter slice: state crossFilters , actions <code>toggleCrossFilter</code> , <code>setSlicerFilter</code> , <code>clearCrossFilter</code> , <code>clearDashboard</code> , selector <code>selectCrossFilters</code> . Not <code>DashboardFilters</code> , <code>focus</code> or <code>resetAll</code> . Zustand ≥ 4.4 (<code>useShallow</code>). 1 file

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
M09 P3	ADAPT	\$APP/hooks/ useSemanticQuery.ts	\$MANIFEST/hooks/ useSemanticQuery.ts	POST <code>/api/powerbi/query</code> → UDP API endpoint C1 through UDP's axios and auth. Drop useDaxLogStore (PowerReact DAX inspector). Keep the query key <code>["dax", kind, datasetId, dax]</code> , keepPreviousData and the abort signal. 1 file
M10 P3	ADAPT	\$APP/lib/geo/geoAssets.ts, useGeoAsset.ts	\$MANIFEST/geo/	getJSON → UDP axios; <code>/geo/</code> base → where A01 is served. Keep staleTime and gcTime Infinity. 2 files
M11 P3	ADAPT	\$APP/components/manifest/ registry.tsx	\$MANIFEST/components/ registry.tsx	Wrappers from <code>../..UI/powerbi-visuals</code> (R02); ChartCard (slicer frame) → UdpFluentCard; <code>import.meta.env.VITE_MAP_TILE_*</code> → ConfigService (C4). Keep <code>REGISTRY satisfies Record<ManifestComponentName, unknown></code> . 1 file
M12 P3	ADAPT	\$APP/components/manifest/ ManifestVisual.tsx	\$MANIFEST/components/ ManifestVisual.tsx	Imports → M04, M05, M08, M09, M10; errorMessage → UDP error helper. Live and Sample stay separate components (hooks never change order). 1 file
M13 P3	ADAPT	\$APP/components/manifest/ ManifestDashboard.tsx + manifest.css	\$MANIFEST/components/	ReportTemplate → UDP page layout; RemovableChip → UDP chip; 4 Tailwind class lists → UDP styling. The data area must set <code>container-type: inline-size</code> (grid and elements respond to it). <code>var(--u-gap)</code> → <code>var(--u-gap, var(--spacingHorizontalL))</code> . 2 files
M14 P3	ADAPT	\$APP/components/manifest/ ManifestFilterBar.tsx	\$MANIFEST/components/ ManifestFilterBar.tsx	Imports as M12; 1 Tailwind class list. 1 file
M15 P3	REPLACE	\$APP/components/manifest/ ManifestSlicer.tsx	\$MANIFEST/components/ ManifestSlicer.tsx	Re-implement on UdpTablist (≤ 6 short options) and UdpDropdown (multi-select with search), same props: label, options, selected, mode, multiple, allLabel, onChange, loading, error, testId. 1 file
M16 P3	NEW	–	\$ENT/udp/pages/ UdpManifestReportPage/	Loads a manifest by id (C2), validates it, renders ManifestDashboard. Register <code>UdpManifestReportPage: () ⇒ import('../UdpManifestReportPage')</code> in UdpPagesLoaders. 2 files

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
M17 P4	NEW	–	\$ENT/UI/widgets/ library/ ManifestVisualWidget/	Widget module: default , layout (w, h ranges), params { manifestId, visualId }; renders one manifest visual on a dashboard. Register in UdpPagesLoaders. 2 files
M18 P3	ADAPT	\$APP/pages/ ManifestPreview.tsx	\$ENT/udp/pages/ UdpManifestPreviewPage /	Admin tool: upload (≤ 2 MB), validate, Live / Sample. Replace ReportTemplate, Card, Tabs, ErrorNote, LoadingState, useThemeStore, getJson and lucide-react icons with UDP equivalents (20 Tailwind class lists). 2 files
M19 P3	REPLACE	\$APP/api/http.ts	UDP axios / useAxiosGet	getJson, postJson, errorMessage. 1 file
M20 P3	REPLACE	\$APP/store/theme.ts	UDP shell theme	A manifest's theme field maps to the UDP theme or is ignored. 1 file
M21 P3	REPLACE	\$APP/components/template/*, \$APP/components/ui/ {Card, InfoCurtain, Popover, p rimitives, cx}	UDP layout and components	PowerReact chrome: page frame, cards, popover, tabs, chips. Use UDP's. 7 files

6 W3 • Assets and scripts

5 items · 14 files · Map boundaries, seed manifests, JSON Schema, optional Univerus theme.

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
A01 P3	COPY	public/geo/{index.json, world-110m.topo.json, na- admin1.topo.json, na- tiles.json, README.md}	\$STATIC/geo/	Natural Earth, public domain (about 320 KB). Served from the app origin; no external map hosts. 5 files
A02 P3	COPY	scripts/build-geo.mjs	packages/udp-react- enterprise-component- library/scripts/build- geo.mjs	Rebuilds A01; devDependencies topojson-server, topojson-simplify, topojson-client. Writes to the A01 location (edit OUT). 1 file
A03 P3	COPY	public/manifests/{index, sample-manifest, condition- works, delivery-lifecycle, locations-performance}.json	\$MANIFEST/ __fixtures__/ manifests/ and the C2 seed	Illustrative samples. dataSource ids point to the demo model: replace per tenant. 5 files
A04 P3	ADAPT	scripts/generate-manifest- schema.ts → public/ manifests/ manifest.schema.json	packages/udp-react- enterprise-component- library/scripts/ + published schema	Regenerate the JSON Schema from M02; publish it where Univerus-Lens reads it. A test keeps it in sync. 2 files

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
A05 P3	COPY	\$APP/theme/tokens.css	\$ENT/styles/univerus-theme.css	Optional: the <code>--u-*</code> roles of Neo-Glass / Nocturne, loaded only on pages that want the approved Univerus look; with fonts <code>@fontsource-variable/space-grotesk</code> and <code>/montserrat</code> . Without it the bridge uses Fluent tokens. 1 file

7 W4 • UDP API (outside the monorepo)

3 items · 3 files · Query endpoint and manifest store, built by the UDP API team.

ID	ACTION	SOURCE (POWERREACT)	DESTINATION (UDP)	NOTES
S01 P2	REFERENCE	server/routes/powerbi.ts (POST /powerbi/query), server/services/powerbi.ts (executeDaxQuery), server/config.ts (allowedDatasets)	UDP API: endpoint C1	Contract in section 7. The UDP team implements it; the agent does not edit the API repo. 3 files
S02 P3	NEW	–	UDP API: endpoint C2 (manifest store)	Proposed contract in section 7.
S03 P2	NEW	–	ConfigService keys (C4)	Allowed datasets per tenant; optional map tile URL template + attribution (Azure Maps via AZURE_MAP_SUBSCRIPTION_KEY).

C1 • DAX query endpoint (contract)

```
POST {tenantV2ApiUrl}/powerbi/query
Authorization: Bearer <UDP user token>

Request { "query": "EVALUATE ...",           // one top-level EVALUATE (schema-checked)
          "workspaceId": "<uuid>",
          "datasetId": "<uuid>" }           // must be in the tenant's allow-list

200 { "success": true, "rows": [ { "table[Column]": ..., "[Measure]": ... } ],
     "rowCount": 12, "executionTimeMs": 140 }
403 { "success": false, "error": "Dataset ... is not allowed", "hint": "..." }
4xx/5xx { "success": false, "error": "DAX Query Execution failed", "hint": "...", "details": ... }

Upstream: POST https://api.powerbi.com/v1.0/myorg/groups/{workspaceId}/datasets/{datasetId}/executeQueries
          { "queries": [ { "query": ... } ], "serializerSettings": { "includeNulls": true },
            "impersonatedUserName": "<UPN>" } // only for row-level-security models

Limits: 100,000 rows or 1,000,000 values, 15 MB, one table per query;
        120 requests / minute / identity → cache by DAX text, per-user tokens.
```

C2 • Manifest store (proposed contract)

```
GET {tenantV2ApiUrl}/manifests           → [ { id, title, version, updatedAt } ]
GET {tenantV2ApiUrl}/manifests/{id}?version= → the manifest JSON
PUT {tenantV2ApiUrl}/manifests/{id}       → validates against manifest.schema.json,
                                           stores a new version, returns { id, version }

Scope: per tenant. Keep the previous version for rollback.
```

8 Invariants (never change)

- Tags `udp-pbi-*`, classes `UdpPbi*`, event names (`dataPointClick`, `exportData`, `focusModeChange`, `viewChange`) and every public prop name.
- Manifest component names `Univerus*` and `schemaVersion: 1`: they are the contract with Univerus-Lens.
- Components are presentational: they never fetch, never build DAX. Data in as props, events out.
- Colours only through `--pbi-*` roles; `--u-*` is read only in `tokens-bridge.css`; no literal colours.
- Filters reach DAX only through `injectCrossFilters` (one `TREATAS` per filter); a clicked value is never concatenated into a query.
- Destination layout mirrors `$PKG` under `$STENCIL`, so relative imports and `styleUrls` stay valid.
- No secrets, `.env` files or tenant data move. Seed manifests use demo ids only.

9 Dependencies

Add these where listed. Versions are the ones PowerReact builds and tests with.

PACKAGE	VERSION	ADD TO	STATUS IN UDP TODAY
d3-array	^3.2.4	\$STENCIL_PKG	New
d3-contour	^4.0.2	\$STENCIL_PKG	New
d3-geo	^3.1.1	\$STENCIL_PKG	New
d3-hexbin	^0.2.2	\$STENCIL_PKG	New
d3-hierarchy	^3.1.2	\$STENCIL_PKG	New
d3-scale	^4.0.2	\$STENCIL_PKG	New
d3-selection	^3.0.0	\$STENCIL_PKG	New
d3-shape	^3.2.0	\$STENCIL_PKG	New
d3-time	^3.1.0	\$STENCIL_PKG	New
d3-transition	^3.0.1	\$STENCIL_PKG	New
d3-zoom	^3.0.0	\$STENCIL_PKG	New
topojson-client	^3.1.0	\$STENCIL_PKG	New
lucide	^1.52.0	\$STENCIL_PKG	New, unless V11 maps the icons to UDP's registry
xlsx	0.18.5 (UDP's)	\$STENCIL_PKG	Present; keep it
@types/d3-* (11), @types/geojson, @types/topojson-client	as in PowerReact	\$STENCIL_PKG (dev)	New
zod	^4.6.5	\$ENT package	New
zustand	≥ 4.4 (UDP: ^4.3.7)	\$ENT package	Present; confirm ≥ 4.4 for useShallow
@tanstack/react-query	^5	\$ENT package	Present
@types/geojson, @types/topojson-specification	latest	\$ENT package (dev)	New
topojson-server, topojson-simplify, topojson-client	^3	\$ENT package (dev, A02)	New

10 Phases, gates and acceptance checks

A phase is done only when every check of its gate passes. P1 and P2 run in parallel.

PHASE	ITEMS	GATE	CHECKS
P1 · Visual library	V01–V16, R01–R03	G1 · Foundations ready (library)	☐ pnpm --filter @univerus/udp-stencil-component-library build → succeeds; generated docs list 25 tags udp-pbi-* ☐ pnpm --filter @univerus/udp-stencil-component-library test:unit → layout, util, SSR and token tests pass ☐ grep -r "var(--u-" \$STENCIL → matches only styles/tokens-bridge.css ☐ Token guard tests (V15) pass: every --pbi-* bridged with a Fluent fallback; no literal colours in \$STENCIL ☐ Storybook: every family renders in Fluent light and body.dark-theme ☐ pnpm --filter @univerus/udp-react-enterprise-component-library build → the 25 UdpPbi* wrappers are exported
P2 · Query endpoint	S01, S03 (UDP API team)	G1 · Foundations ready (data)	☐ POST C1 with an allowed dataset → 200 and rows ☐ POST C1 with another dataset → 403 ☐ Row-level-security model → rows scoped to the signed-in user
P3 · Manifest reader	M01–M16, M18–M21, A01–A05, S02	G2 · Pilot signed off	☐ Manifest unit tests (M06) pass; the regenerated JSON Schema equals PowerReact's public/manifests/manifest.schema.json ☐ UdpManifestReportPage renders condition-works in Sample mode with no network request ☐ Live: each visual issues one C1 call; a click re-queries the others with TREATASC in the DAX ☐ Maps load /geo/*.topo.json from the app origin only ☐ One Lens report runs for a pilot tenant and is signed off
P4 · Rollout	M17	G3 · All tenants	☐ ManifestVisualWidget placed on a dashboard; reports published through C2 for every tenant

11 Open decisions

Each has a proposed answer the agent may use until the UDP team decides.

ID	QUESTION	PROPOSED ANSWER
D1	Unit-test runner for the enterprise library (its test script is lint + build today)	Add Vitest with a test:unit script (the manifest tests are Vitest)
D2	Icons: lucide or UDP's icon registry (V11)	UDP registry if it covers the 38 icons; else add lucide
D3	Identity for C1: user token or service principal	User token (MSAL) for row-level-security or SSO models; service principal elsewhere
D4	Where manifests live (C2) and who publishes	UDP API + database, per tenant, versioned; Lens publishes
D5	Static path for /geo in each host app (A01)	The host app's public/geo/

ID	QUESTION	PROPOSED ANSWER
D6	Adopt <code>univerus-theme.css</code> (A05) for the approved Univerus look	Yes on report pages (<code>frame="card"</code>); Fluent fallback elsewhere

12 Not moving

Anything not listed in sections 4–7 stays in PowerReact. In particular:

- **EXCLUDE** PowerReact server (`server/**`): replaced by C1 and C2 in the UDP API.
- **EXCLUDE** App shell and pages unrelated to manifests: `src/components/{briefing,chat,embed,layout,visuals}` , `src/pages/*` except ManifestPreview (M18) and Gallery (content only, V13).
- **EXCLUDE** `src/store/daxLog.ts` (DAX inspector) and `src/store/filters.ts` beyond the cross-filter slice (M08).
- **EXCLUDE** Build and tooling of PowerReact: `stencil.config.ts` , Vite, Tailwind, Playwright config, `index.html` , `packages/udp-powerbi-visuals/{index.js, components/, hydrate/, dist/, docs/}` .
- **EXCLUDE** Agent skills in `.agents/` and `.claude/` (PowerReact-specific; UDP keeps its own `ai-skills`).
- **EXCLUDE** `.env` , credentials and any tenant data.

A Appendix • machine-readable move list

The items of sections 4–7 as JSON, one per line (full twin with variables, gates and contracts: `docs/udp-migration/udp-migration-manifest.json`).

```
[
{"id":"V01","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/components/kpi/udp-pbi-
*/","destination":"$STENCIL/components/kpi/","files":15,"notes":"5 components: kpi-card, kpi-hero, kpi-trend, kpi-bullet, kpi-
variance (tsx, css, readme.md each). Keep the mirrored layout: relative imports `../../styles/|utils|functional` must resolve
to $STENCIL."},
{"id":"V02","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/components/charts/udp-pbi-
*/","destination":"$STENCIL/components/charts/","files":54,"notes":"18 components: boxplot, bullet-bars, calendar-heatmap,
choropleth, column-chart, diverging-bars, donut, dot-plot, ibcs-variance, point-map, ranking-bars, scatter, spotlight-bars,
stacked-bars, timeline, treemap, trend-chart, waterfall."},
{"id":"V03","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/components/tables/udp-pbi-
*/","destination":"$STENCIL/components/tables/","files":6,"notes":"2 components: data-table, matrix."},
{"id":"V04","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/functional/*.tsx","destination":"$STENCIL/functional/","f
iles":7,"notes":"chart-kit, frame, icon, kpi-shell, map-kit, meter, states."},
{"id":"V05","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/utils/** (non-
test)","destination":"$STENCIL/utils/","files":30,"notes":"Root (data, events, formats, icons, palette, roving, series-chart,
width-observer), export/ (csv, download, xlsx), layout/ (17 modules incl. geo, kpi), table/ (builders, model)."},
{"id":"V06","phase":1,"workstream":"W1","action":"COPY","source":"$PKG/styles/*.css","destination":"$STENCIL/styles/","files":7,
"notes":"tokens-bridge, motion, shadow, surfaces, charts, kpi, maps. tokens-bridge.css stays the only file that reads `--u-
*`. "},
{"id":"V07","phase":1,"workstream":"W1","action":"ADAPT","source":"$PKG/utils/**/*test.ts","destination":"$STENCIL/utils/ (same
folders)","files":8,"notes":"Vitest → Stencil Jest (`stencil test --spec`): drop `import ... from 'vitest'` (globals). vocabulary-
e.test.ts reads `public/geo/*.json`: point it at the copy made in A01 (or a test fixture)."},
{"id":"V08","phase":1,"workstream":"W1","action":"REGENERATE","source":"$PKG/components.d.ts","destination":"$STENCIL_PKG/src/co
mponents.d.ts","files":1,"notes":"UDP's Stencil build writes its own; do not copy."},
{"id":"V09","phase":1,"workstream":"W1","action":"ADAPT","source":"$PKG/index.ts","destination":"$STENCIL_PKG public type
entry","files":1,"notes":"Add its type re-exports (events, formats, table model, KpiIconName, VisualFrameMode, BarMeta) to the
library's public types."},
{"id":"V10","phase":1,"workstream":"W1","action":"ADAPT","source":"packages/udp-powerbi-
visuals/package.json","destination":"$STENCIL_PKG/package.json","files":1,"notes":"Merge dependencies per section 9. Keep UDP's
xlsx 0.18.5 (do not add the 0.20 tarball)."},
{"id":"V11","phase":1,"workstream":"W1","action":"ADAPT","source":"$PKG/utils/icons.ts","destination":"$STENCIL/utils/icons.ts",
"files":1,"notes":"Uses `lucide` (38 icons). Either add `lucide` (section 9) or map each icon to UDP's icon registry; keep the
exported names (KPI_ICONS, UI_ICONS, KpiIconName) – the manifest schema lists the KPI icon names."},
{"id":"V12","phase":1,"workstream":"W1","action":"COPY","source":"$APP/components/powerbi-
visuals/_fixtures_/samples.ts","destination":"$STENCIL/_fixtures_/samples.ts","files":1,"notes":"Sample data (illustrative,
no client data). Rewrite the type import `@powerreact/udp-powerbi-visuals` → relative imports of the same types."},
{"id":"V13","phase":1,"workstream":"W1","action":"NEW","source":"$APP/pages/Gallery.tsx (reference for
content)","destination":"$STENCIL/stories/<family>.stories.ts","files":11,"notes":"Replaces the Gallery page: one Storybook file
per family (KPI, Ranking, Magnitude, Change over time, Deviation, Part-to-whole, Flow, Distribution, Correlation, Spatial, Exact
values) using V12 data; a toolbar toggle for UDP dark (`body.dark-theme`) and the Univerus look (A05)."},
{"id":"V14","phase":1,"workstream":"W1","action":"ADAPT","source":"$APP/components/powerbi-
visuals/elements.ssr.test.ts","destination":"$STENCIL/_tests_/elements.spec.tsx","files":1,"notes":"Server-render smoke tests
(full, empty, frame=`none`) for all 25 components; port to Stencil `newSpecPage` or keep the hydrate renderer."},
{"id":"V15","phase":1,"workstream":"W1","action":"ADAPT","source":"$APP/theme/tokensBridge.test.ts + package part of
$APP/theme/noRawColors.test.ts","destination":"$STENCIL/_tests_/tokens.spec.ts","files":2,"notes":"Guards: every `--pbi-*`
read exists in the bridge with a Fluent fallback; no `var(--u-` outside tokens-bridge.css; no hex / rgb / hsl literals in
$STENCIL."},
{"id":"V16","phase":1,"workstream":"W1","action":"ADAPT","source":"packages/udp-powerbi-visuals/README.md,
MIGRATION.md","destination":"packages/udp-react-enterprise-component-library/docs/charts/powerbi-
visuals.md","files":2,"notes":"Component table, contract, token bridge, maps, the documented `frame=``card``` deviation."},
{"id":"R01","phase":1,"workstream":"W1","action":"REGENERATE","source":"$APP/components/powerbi-
visuals/generated/components.ts","destination":"$WRAP_PKG","files":1,"notes":"Built by UDP's React output target (React bundle).
Never copy PowerReact's generated file."},
{"id":"R02","phase":1,"workstream":"W1","action":"NEW","source":"-","destination":"$ENT/UI/powerbi-visuals/UdpPbi<Name>.tsx +
index.ts","files":26,"notes":"One thin wrapper per component (25) + barrel, the pattern of UDP's chart wrappers; export the
barrel from $ENT/index.ts"},
{"id":"R03","phase":1,"workstream":"W1","action":"ADAPT","source":"$APP/components/powerbi-
visuals/index.ts","destination":"$ENT/UI/powerbi-visuals/index.ts","files":1,"notes":"Public type re-exports used by the
manifest code (DataPointClickDetail, DataPointValue, GeoPoint, RegionValue, KpiPoint, ...), now from the UDP wrapper package."},
{"id":"M01","phase":3,"workstream":"W2","action":"COPY","source":"$APP/lib/manifest/schema.ts","destination":"$MANIFEST/lib/sche
ma.ts","files":1,"notes":"Zod 4 schema: 26 components, `schemaVersion: 1`, strict objects."},
{"id":"M02","phase":3,"workstream":"W2","action":"COPY","source":"$APP/lib/manifest/jsonSchema.ts","destination":"$MANIFEST/lib/
jsonSchema.ts","files":1,"notes":"Set MANIFEST_SCHEMA_FILE to the UDP location of the published JSON Schema (A04)."},
{"id":"M03","phase":3,"workstream":"W2","action":"COPY","source":"$APP/lib/manifest/daxInjection.ts","destination":"$MANIFEST/li
b/daxInjection.ts","files":1,"notes":"Import `../dax/daxBuilder` → `../dax` (M07). The only place filters enter a query."},
{"id":"M04","phase":3,"workstream":"W2","action":"COPY","source":"$APP/lib/manifest/mapRows.ts","destination":"$MANIFEST/lib/map
Rows.ts","files":1,"notes":"Imports: `@powerreact/udp-powerbi-visuals` (types) → `@univerus/udp-react-stencil-component-
library`; `../dax/parse`, `../dax/types` → `../dax`. "},
{"id":"M05","phase":3,"workstream":"W2","action":"COPY","source":"$APP/lib/manifest/crossFilters.ts","destination":"$MANIFEST/li
b/crossFilters.ts","files":1,"notes":"Same import rewrites as M04. Holds Sample mode and which filters each visual obeys."},
{"id":"M06","phase":3,"workstream":"W2","action":"ADAPT","source":"$APP/lib/manifest/*.test.ts +
$APP/components/manifest/registry.test.ts","destination":"$MANIFEST/_tests_/","files":7,"notes":"Needs a unit-test runner in
the enterprise library (decision D1). Bundled manifests (A03) as fixtures."},
{"id":"M07","phase":3,"workstream":"W2","action":"EXTRACT","source":"$APP/lib/dax/daxBuilder.ts, parse.ts,
types.ts","destination":"$MANIFEST/lib/dax.ts","files":1,"notes":"Only: daxString, DaxLiteralValue, daxLiteral, treatAs, call
(daxBuilder); toNumber, toLabel (parse); DaxRow (types) – with their test cases. Not the asset-page builders, GROUP_COLUMN or
CLASS_COLUMN."},
{"id":"M08","phase":3,"workstream":"W2","action":"EXTRACT","source":"$APP/store/filters.ts","destination":"$MANIFEST/store/manif
estFilters.ts","files":1,"notes":"Only the cross-filter slice: state `crossFilters`, actions toggleCrossFilter, setSlicerFilter,
clearCrossFilter, clearDashboard, selector selectCrossFilters. Not DashboardFilters, focus or resetAll. Zustand ≥ 4.4
(useShallow)."},
{"id":"M09","phase":3,"workstream":"W2","action":"ADAPT","source":"$APP/hooks/useSemanticQuery.ts","destination":"$MANIFEST/hook
s/useSemanticQuery.ts","files":1,"notes":"POST `/api/powerbi/query` → UDP API endpoint C1 through UDP's axios and auth. Drop
```



```

useDaxLogStore (PowerReact DAX inspector). Keep the query key `[\"dax\", kind, datasetId, dax]`, keepPreviousData and the abort
signal.\"},
{\"id\":\"M10\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/lib/geo/geoAssets.ts,
useGeoAsset.ts\",\"destination\":\"$MANIFEST/geo/\",\"files\":2,\"notes\":\"getJSON → UDP axios; `geo/` base → where A01 is served. Keep
staleTime and gcTime Infinity.\"},
{\"id\":\"M11\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/components/manifest/registry.tsx\",\"destination\":\"$MANIFE
ST/components/registry.tsx\",\"files\":1,\"notes\":\"Wrappers from `../UI/powerbi-visuals` (R02); ChartCard (slicer frame) →
UdpFluentCard; `import.meta.env.VITE_MAP_TILE_*` → ConfigService (C4). Keep `REGISTRY` satisfies Record<ManifestComponentName,
unknown>\"},
{\"id\":\"M12\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/components/manifest/ManifestVisual.tsx\",\"destination\":\"$
MANIFEST/components/ManifestVisual.tsx\",\"files\":1,\"notes\":\"Imports → M04, M05, M08, M09, M10; errorMessage → UDP error helper.
Live and Sample stay separate components (hooks never change order).\"},
{\"id\":\"M13\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/components/manifest/ManifestDashboard.tsx +
manifest.css\",\"destination\":\"$MANIFEST/components/\", \"files\":2,\"notes\":\"ReportTemplate → UDP page layout; RemovableChip → UDP
chip; 4 Tailwind class lists → UDP styling. The data area must set `container-type: inline-size` (grid and elements respond to
it). `var(--u-gap)` → `var(--u-gap, var(--spacingHorizontal))`\"},
{\"id\":\"M14\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/components/manifest/ManifestFilterBar.tsx\",\"destination\"
: \"$MANIFEST/components/ManifestFilterBar.tsx\", \"files\":1,\"notes\":\"Imports as M12; 1 Tailwind class list.\"},
{\"id\":\"M15\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"REPLACE\",\"source\":\"$APP/components/manifest/ManifestSlicer.tsx\", \"destination\":
\"$MANIFEST/components/ManifestSlicer.tsx\", \"files\":1,\"notes\":\"Re-implement on UdpTablist (≤ 6 short options) and UdpDropdown
(multi-select with search), same props: label, options, selected, mode, multiple, allLabel, onChange, loading, error, testId.\"},
{\"id\":\"M16\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"NEW\",\"source\":\"-\", \"destination\":\"$ENT/udp/pages/UdpManifestReportPage/\", \"files
\":2,\"notes\":\"Loads a manifest by id (C2), validates it, renders ManifestDashboard. Register `UdpManifestReportPage: () ⇒
import(`../UdpManifestReportPage`)` in UdpPagesLoaders.\"},
{\"id\":\"M17\",\"phase\":4,\"workstream\":\"W2\",\"action\":\"NEW\",\"source\":\"-\", \"destination\":\"$ENT/UI/widgets/library/ManifestVisualWidget/
\", \"files\":2,\"notes\":\"Widget module: `default`, `layout` (w, h ranges), `params` { manifestId, visualId }; renders one manifest
visual on a dashboard. Register in UdpPagesLoaders.\"},
{\"id\":\"M18\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"ADAPT\",\"source\":\"$APP/pages/ManifestPreview.tsx\", \"destination\":\"$ENT/udp/pages
/UdpManifestPreviewPage/\", \"files\":2,\"notes\":\"Admin tool: upload (≤ 2 MB), validate, Live / Sample. Replace ReportTemplate, Card,
Tabs, ErrorNote, LoadingState, useThemeStore, getJson and lucide-react icons with UDP equivalents (20 Tailwind class lists).\"},
{\"id\":\"M19\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"REPLACE\",\"source\":\"$APP/api/http.ts\", \"destination\":\"UDP axios /
useAxiosGet\", \"files\":1,\"notes\":\"getJSON, postJson, errorMessage\"},
{\"id\":\"M20\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"REPLACE\",\"source\":\"$APP/store/theme.ts\", \"destination\":\"UDP shell
theme\", \"files\":1,\"notes\":\"A manifest's `theme` field maps to the UDP theme or is ignored.\"},
{\"id\":\"M21\",\"phase\":3,\"workstream\":\"W2\",\"action\":\"REPLACE\",\"source\":\"$APP/components/template/*,
$APP/components/ui/{Card,InfoCurtain,Popover,primitives,cx}\"\", \"destination\":\"UDP layout and
components\", \"files\":7,\"notes\":\"PowerReact chrome: page frame, cards, popover, tabs, chips. Use UDP's.\"},
{\"id\":\"A01\",\"phase\":3,\"workstream\":\"W3\",\"action\":\"COPY\",\"source\":\"public/geo/{index.json, world-110m.topo.json, na-
admin1.topo.json, na-tiles.json, README.md}\"\", \"destination\":\"$STATIC/geo/\", \"files\":5,\"notes\":\"Natural Earth, public domain (about
320 KB). Served from the app origin; no external map hosts.\"},
{\"id\":\"A02\",\"phase\":3,\"workstream\":\"W3\",\"action\":\"COPY\",\"source\":\"scripts/build-geo.mjs\", \"destination\":\"packages/udp-react-
enterprise-component-library/scripts/build-geo.mjs\", \"files\":1,\"notes\":\"Rebuilds A01; devDependencies topojson-server, topojson-
simplify, topojson-client. Writes to the A01 location (edit OUT).\"},
{\"id\":\"A03\",\"phase\":3,\"workstream\":\"W3\",\"action\":\"COPY\",\"source\":\"public/manifests/{index, sample-manifest, condition-works,
delivery-lifecycle, locations-performance}.json\", \"destination\":\"$MANIFEST/_fixtures_/manifests/ and the C2
seed\", \"files\":5,\"notes\":\"Illustrative samples. `dataSource` ids point to the demo model: replace per tenant.\"},
{\"id\":\"A04\",\"phase\":3,\"workstream\":\"W3\",\"action\":\"ADAPT\",\"source\":\"scripts/generate-manifest-schema.ts →
public/manifests/manifest.schema.json\", \"destination\":\"packages/udp-react-enterprise-component-library/scripts/ + published
schema\", \"files\":2,\"notes\":\"Regenerate the JSON Schema from M02; publish it where Univerus-Lens reads it. A test keeps it in
sync.\"},
{\"id\":\"A05\",\"phase\":3,\"workstream\":\"W3\",\"action\":\"COPY\",\"source\":\"$APP/theme/tokens.css\", \"destination\":\"$ENT/styles/univerus-
theme.css\", \"files\":1,\"notes\":\"Optional: the `--u-*` roles of Neo-Glass / Nocturne, loaded only on pages that want the approved
Univerus look; with fonts @fontsource-variable/space-grotesk and /montserrat. Without it the bridge uses Fluent tokens.\"},
{\"id\":\"S01\",\"phase\":2,\"workstream\":\"W4\",\"action\":\"REFERENCE\",\"source\":\"server/routes/powerbi.ts (POST /powerbi/query),
server/services/powerbi.ts (executeDaxQuery), server/config.ts (allowedDatasets)\", \"destination\":\"UDP API: endpoint
C1\", \"files\":3,\"notes\":\"Contract in section 7. The UDP team implements it; the agent does not edit the API repo.\"},
{\"id\":\"S02\",\"phase\":3,\"workstream\":\"W4\",\"action\":\"NEW\",\"source\":\"-\", \"destination\":\"UDP API: endpoint C2 (manifest
store)\", \"files\":0,\"notes\":\"Proposed contract in section 7.\"},
{\"id\":\"S03\",\"phase\":2,\"workstream\":\"W4\",\"action\":\"NEW\",\"source\":\"-\", \"destination\":\"ConfigService keys
(C4)\", \"files\":0,\"notes\":\"Allowed datasets per tenant; optional map tile URL template + attribution (Azure Maps via
AZURE_MAP_SUBSCRIPTION_KEY).\"}
]

```